Assignment 4

Q1

[0.60 points]

Use the Pumping Lemma to show that the following languages are not regular:

a. $A = \{1^n 2^n 3^n \mid n \geq 0\}$
b. $B = \{www \mid w \in \{a, b\}^* \}$
c. $C = \{a^{2^n} \mid n \geq 0\}$ (Here a^{2^n} means a string of 2^n a's)

A1

a) $A = \{1^n 2^n 3^n \mid n \geq 0\}$

Proof.

Assume that A is regular. Let p be the pumping length given by the pumping lemma. Choose s to be the string $1^p 2^p 3^p$. Because $s \in A$ and $|s| \geq p$, the pumping lemma guarantees that s can be split into three pieces, $s = xyz$, where $|xy| \leq p$, $y \neq \varepsilon$, and for any $i \geq 0$ the string $xy^i z \in A$.

Since $|xy| \leq p$, x and y can only contain 1s. So, $xyyz$ will not have equal number of 1s, 2s and 3s. Hence $xyyz$ is not a member of A, a contradiction. Therefore, A is not regular.

b) $B = \{www \mid w \in \{a, b\}^* \}$

Proof.

Assume that B is regular. Let p be the pumping length given by the pumping lemma. Choose s to be the string $a^p b a^p b a^p b$. Because $s \in B$ and $|s| \geq p$, the pumping lemma guarantees that s can be split into three pieces, $s = xyz$, where $|xy| \leq p$, $y \neq \varepsilon$, and for any $i \geq 0$ the string $xy^i z \in B$.

Since $|xy| \leq p$, x and y can only contain as. Clearly, $xyyz \notin B$, a contradiction. Therefore, B is not regular.

c) $C = \{a^{2^n} \mid n \geq 0\}$ (Here a^{2^n} means a string of 2^n a's)

Proof.

Assume that C is regular. Let p be the pumping length given by the pumping lemma. Choose s to be the string a^{2^p}. Because $s \in C$ and $|s| \geq p$, the pumping lemma guarantees that s can be split into three pieces, $s = xyz$, where $|xy| \leq p$, $y \neq \varepsilon$, and for any $i \geq 0$ the string $xy^i z \in C$.

Since $|xy| \leq p$ and $y \neq \varepsilon$, we have $|y| \leq p$. Thus, $2^p < |xyyz| \leq p + 2^p < 2^{p+1}$. Hence $xyyz$ is not a member of C, a contradiction. Therefore, C is not regular.

Q2

[0.20 points]

Describe an error in the following "proof that 0^*1^* is not a regular language. An error must exist as 0^*1^* is obviously regular.

"Proof" by contradiction: *Assume that 0^*1^* is regular. Let p be the pumping length for 0^*1^*given the pumping lemma. Choose s to be the string 0^p1^p. You know that s is a member of 0^*1^* but Example 1 in the lecture slides show that s cannot be pumped. Thus you have a contradiction and 0^*1^* is not regular.*

Example 1 of Proving Non-regularity

Pumping Lemma: For every regular language A, there is a p such that if $s \in A$ and $|s| \geq p$ then $s = xyz$ where

1) $xy^iz \in A$ for all $i \geq 0$ $\quad y^i = yy\cdots y$
2) $y \neq \varepsilon$
3) $|xy| \leq p$

Let $D = \{0^k1^k | \, k \geq 0\}$

Show: D is not regular

Proof by Contradiction:

Assume (to get a contradiction) that D is regular.

The pumping lemma gives p as above. Let $s = 0^p1^p \in D$. ($|s| \geq p$ and $s \in A$)

Pumping lemma says that we can divide $s = xyz$ satisfying the 3 conditions.

1. y consists of only 0s => $xyyz$ has excess 0s and thus $xyyz \notin D$
2. y consists of only 1s => opposite contradiction
3. y consists of both 0s and 1s => they will be out of order

$$s = \underbrace{000\cdots000111\cdots111}_{x \quad y \quad z}$$

$\leftarrow \leq p \rightarrow$

All ways of splitting s contradict the pumping lemma. Therefore our assumption (D is regular) is false. We conclude that D is not regular.

Note: Cases 2. and 3. can be eliminated by considering Condition 3) of the Pumping Lemma.

A2

In example 1, the language that we discussed was $D = \{0^k1^k \mid k \geq 0\}$. In example 1, we proved by contradiction through showing that y consists of only 0s and $xyyz$ is not in D (case 1 in the slide). But in this question, we are discussing 0^*1^*, and $xyyz$ can still be expressed as 0^*1^*, so we do not have any contradiction here.

Q3

[1 point]

Prove that the following languages are not regular. You may use the pumping lemma and the closure of the class of regular languages under union, intersection and complement.

a. $\{0^n1^m0^n \mid m, n \geq 0\}$
b. $\{0^m1^n \mid m \neq n\}$

A3

a) $\{0^n1^m0^n \mid m, n \geq 0\}$

Proof 1.

Denote the set by A. Assume that A is regular. let p be the pumping length. Then consider $s = 0^p10^p$. Because $s \in A$ and $|s| \geq p$, the pumping lemma guarantees that s can be split into three pieces, $s = xyz$, where $|xy| \leq p$, $y \neq \varepsilon$, and for any $i \geq 0$ the string $xy^iz \in A$.

Since $|xy| \leq p$, x and y can only contain 0s. So $xyyz = 0^{p+|y|}10^p \notin A$, a contradiction. Thus A is not regular.

Proof 2.

Observe that $A \cap (0^*10^*) = \{0^n10^n \mid n \geq 0\}$. Then we just need to show that $\{0^n10^n \mid n \geq 0\}$ is not regular, so A can not be regular (otherwise it will lead to a contradiction). The rest of the proof is the same as proof 1.

b) $\{0^m1^n \mid m \neq n\}$

Proof 1.

Denote the set by B. Observe that $\bar{B} \cap (0^*1^*) = \{0^k1^k \mid k \geq 0\}$. If B is regular, then $\bar{B}$ must be regular and so would $\bar{B} \cap (0^*1^*)$. But we already proved that $\{0^k1^k \mid k \geq 0\}$ is not regular, so B can not be regular.

Proof 2.

We can prove B to be nonregular by using the pumping lemma directly, though doing so is trickier. Assume that B is regular. Let p be the pumping length given by the pumping lemma. Observe that $p!$ is divisible by all integers from 1 to p, where $p! = p(p-1)\ldots 1$. The string $s = 0^p1^{p+p!} \in B$ and $|s| \geq p$. Thus the pumping lemma implies that s can be divided as xyz with $x = 0^a$, $y = 0^b$ and $z = 0^c1^{p+p!}$ where $b \geq 1$ and $a + b + c = p$. Let s' be the string $xy^{i+1}z$ where $i = p!/b$. Then $y^i = 0^{p!}$ so $y^{i+1} = 0^{b+p!}$, and so $s' = 0^{a+b+c+p!}1^{p+p!}$. That gives $s' = 0^{p+p!}1^{p+p!} \notin B$, a contradiction.

Q4

[0.50 points]

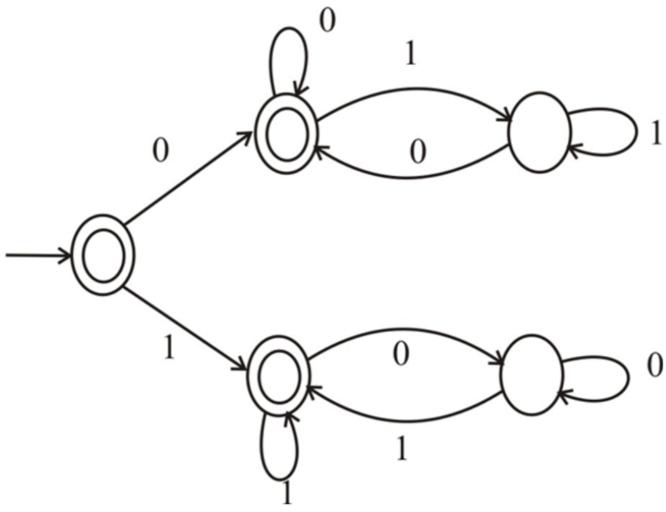
Let $\Sigma = \{0,1\}$ and let

$D = \{w | w$ contains an equal number of occurrences of the substrings 01 and 10$\}$.

Thus $101 \in D$ because 101 contains a single 01 and a single 10, but $1010 \notin D$ because 1010 contains two 10s and one 01. Show that D is a regular language.

A4

The following DFA recognizes the language D, thus D is regular.



Q5

[1 point]

a. Let $B = \{1^k y \mid y \in \{0,1\}^* \text{ and } y \text{ contains at least } k \text{ 1s, for } k \geq 1\}$. Show that B is a regular language.

b. Let $C = \{1^k y \mid y \in \{0,1\}^* \text{ and } y \text{ contains at most } k \text{ 1s, for } k \geq 1\}$. Show that C isn't a regular language.

A5

a) Let $B = \{1^k y \mid y \in \{0,1\}^* \text{ and } y \text{ contains at least } k \text{ 1s, for } k \geq 1\}$. Show that B is a regular language.

Proof.

This is equivalently to say, $B = \{1t \mid t \text{ contains at least one } 1\}$. And B can be expressed by the regular expression $10^*1(1 \cup 0)^*$. Therefore B is a regular language.

b) Let $C = \{1^k y \mid y \in \{0,1\}^* \text{ and } y \text{ contains at most } k \text{ 1s, for } k \geq 1\}$. Show that C isn't a regular language.

Proof.

Assume that C is a regular language. Assume p as the pumping length. Consider a string $S = 1^p 0 1^p \in C$.

Using the pumping lemma, S can be written as,

$$S = 1^p 0 1^p = uvw, \text{ such that } |uv| \leq p, v \neq \varepsilon \text{ and } \forall i \geq 0, uv^i w \in C$$

The trick is that we can set $i = 0$. Since $|uv| \leq p$, u and v can only contain 1s. In this case, $uw = 1^{p-|v|} 0 1^p$. One can see that $uw \notin C$, which leads to a contradiction. Therefore, C is not regular.